

Experiment 9

Aim : Distributed File system

Theory :

- File system describes how files are structured, named, accessed, used, protected and implemented.
- Files are used for permanent use of information on secondary storage.
- In distributed systems, files are available on several computers.
- Computers in distributed system can share these physically dispersed files by using distributed file system.
- Service offers particular function to client and it is a software entity running on some machines.
- Server runs service software on single machine. Client process invokes the service through some set of defined operations called as client interface.
- DFS supports the following:
 - 1) Remote Information Sharing
⇒ Any node in the system can transparently access file irrespective of its location.
 - 2) User mobility
⇒ DFS allows the user to work on different

nodes at different times without physically relocating the secondary storage devices.

3) Availability

⇒ DFS keeps multiple copies of file on different nodes. Failure of any node or copy does not effect the operation.

4) Diskless workstations

⇒ DFS provides transparent file accessing capability; hence, economical diskless workstations can be used.

- DFS provides following three types of services:

1) Storage service

⇒ DFS deals with storage and management of space on secondary storage device that is used to store files in file system.

- Files can be stored in blocks. It offers logical view of storage system by allowing operations for storing and accessing the data in them.

2) True file service

⇒ True file service supports operations on individual file. These are creating and deleting files, modifying and accessing data

from files.

3) Name service

- ⇒ Name service offers mapping between files names to file IDs. It is difficult for human to remember file IDs.
- Most file systems use directories to carry out mapping called as directory service.
- This service supports creation and deletion of directories, carrying out adding new file to directory, deleting the existing file from directory, changing name of the file, etc.

• Features :

1) Performance User Mobility

- ⇒ DFS should permit the user to work on different nodes at different times.
- Performance should not affect if user works on node other than his node.

2) Performance

- ⇒ DFS must give performance same as centralized file system. User should not feel the need to place file explicitly to improve performance.

3) Scalability

⇒ The DFS should also supports for growth of nodes and users in network. Such growth should not lead to service loss or performance degradation of the system.

4) High Availability

⇒ DFS should continue to functional, partial failure occurs in one or more components.

- This failure can be communication link failure, node failure and secondary storage failure.

5) High Reliability

⇒ DFS should support for reliability. It should automatically create backup copies of the critical files that can be loss in the failure.

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